

PRACTICE BCQs FOR 1st Year BDS

Important Information Regarding Practice BCQs

- The following BCQs are provided by the Pharmacology Department for practice purposes only.
- BDS 1st year Students are advised not to consider these as important or indicative of examination questions.
- Furthermore, these BCQs contain three options, whereas examination BCQs typically have five options.

GENERAL PHARMACOLOGY

PHARMACODYNAMICS (Prof. Riffat)

1) Pharmacodynamics is best defined as?

- a) Study of what the drug does to the body
- b) Study of drug excretion
- c) Study of drug absorption and metabolism

Answer: a

2) Which of the following best describes signal transduction??

- a) Drug metabolism in the liver
- b) Conversion of extracellular signal into intracellular response
- c) Transport of drugs across membranes

Answer: b

3) Activation of G-protein coupled receptors primarily leads to?

- a) Direct opening of ion channels
- b) DNA transcription in nucleus only
- c) Activation of intracellular second messengers

Answer: c

4) Which of the following best describes tachyphylaxis?

- a) Gradual increase in drug response
- b) Rapid decrease in response after repeated doses
- c) Delayed drug toxicity

Answer: B

5) Spare receptors are defined as?

- a) Receptors that must all be occupied to produce maximal response
- b) Receptors that do not produce any effect
- c) Receptors present in excess of those required to produce a maximal response

Answer: c

6) Receptor desensitization occurs when?

- a) Receptors increase in number
- b) Receptors become less responsive after continuous stimulation
- c) Drug metabolism increases

Answer: B

7) A partial agonist is a drug that?

- a) Produces submaximal response even at full receptor occupancy
- b) Produces maximum response equal to full agonist
- c) Blocks receptor completely

Answer: a

8) A drug that produces an effect opposite to an agonist via a different receptor is called a?

- a) Functional antagonist
- b) Irreversible antagonist
- c) Partial agonist

Answer: a

9) Potency of a drug refers to:

- a) Maximum effect a drug can produce
- b) Amount of drug required to produce a given effect
- c) Duration of drug action

Answer: B

10) The concentration of a drug that produces 50% of its maximal effect is called:

- A) LD_{50}
- B) EC_{50}
- C) ED_{90}

Answer: B

PHARMACOKINETICS (Dr. Sana)

1. Select a phase 2 biotransformation reaction?

- a. Deamination
- b. Sulfate conjugation
- c. Nitro reduction

Answer: b

2. A drug undergoes extensive first-pass metabolism. Which of the following interventions is most likely to increase its bioavailability?

- a. Increasing hepatic enzyme activity
- b. Administering the drug intravenously
- c. Increasing gastric emptying time

Answer: b

3. Which subfamily of cytochrome P-450 is responsible for the highest fraction of clinically Important drug interactions resulting from metabolism?

- a. CYP1A
- b. CYP2A
- c. CYP3A

Answer: c